Class 11 Alphafold

Joshua Khalil (A17784122)

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Background

we saw last data that the main repository for biomolecular structure (the pdb database) only has ~250,000 entries.

UniPortKB has over 200 million entries

Alphafold

In this hands on session we will utilize alphafold to predict protein structure from sequence

Without the aid of such approaches it can take years of expensive laboratory work to determine the structure of just one protein. With AlphaFold we can now accurately compute a typical protein structure in as little as ten minutes.

EBI Alphafold database

The EBI Alphafold database contains lots of computed structure models. It increasingly likely that the structures you are interested in is already in the database. at https://alphafold.ebi.ac.uk/

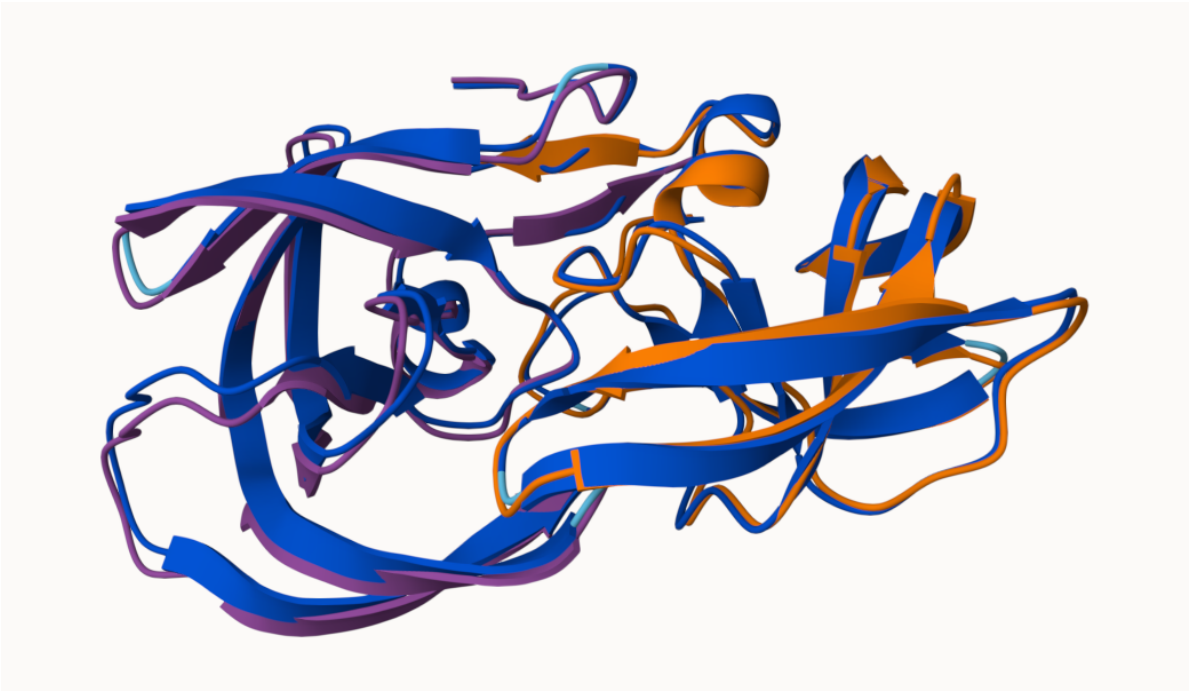
There are three major outputs from alphafold

1. A model of structure in **PDB** format
2. a **pLDDT score**: that tells us how confident the model is for a given residue in your protein (high values are good above 70)
3. a **PAE score** that tels us about protein packing quality.

If you can not find a matching entry for the sequence you are interested in AFDB you can run Alphafold yourself

Running AlphaFold

If you can't find an existing structure of interest in the PDB or AFDB then you can generate your own structure predictions with ColabFold https://colab.research.google.com/github/sokrypton/ColabFold/blob/main/AlphaFold2.ipynb



Custom analysis of resulting models

In this section we will read the results of the more complicated HIV-Pr dimer AlphaFold2 models into R with the help of the Bio3D package. You can do the same thing for the monomer models if you wish but again they will be less interesting as the monomer is not physiologically relevant.

For tidiness we can move our AlphaFold results directory into our RStudio project directory. In this example my results are in the director `results_dir` (set below). You should change this to match your directory/folder name.

```
library(bio3d)
results_dir <- "hivpr_23119"
p <- read.pdb("hivpr_23119/")
pdb_files <- list.files(path=results_dir,
                        pattern="*.pdb",
                        full.names = TRUE)
```

```
pdb_files <-list.files("hivpr_23119", pattern=".pdb", full.names = T)
```

```
pdbs <- pdbaln(pdb_files, fit=TRUE, exefile="msa")
```

```
Reading PDB files:
hivpr_23119/HIVpr_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb
hivpr_23119/HIVpr_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb
hivpr_23119/HIVpr_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb
hivpr_23119/HIVpr_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb
hivpr_23119/HIVpr_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb
.....

Extracting sequences

pdb/seq: 1   name: hivpr_23119/HIVpr_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_
pdb/seq: 2   name: hivpr_23119/HIVpr_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_
pdb/seq: 3   name: hivpr_23119/HIVpr_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_
pdb/seq: 4   name: hivpr_23119/HIVpr_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_
pdb/seq: 5   name: hivpr_23119/HIVpr_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_
```

```
#library(bio3dview)
#view.pdbs(pdbs)
```

How similar or different are my models?

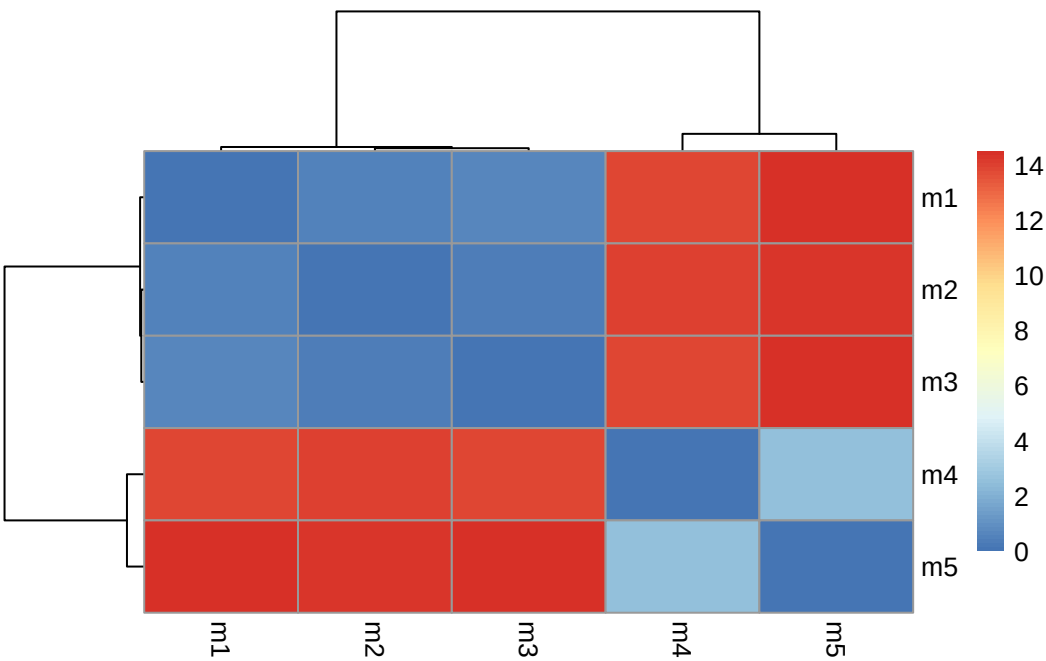
```r
rd <-rmsd(pdbs, fit=T)
```

```
Warning in rmsd(pdbs, fit = T): No indices provided, using the 198 non NA positions
```

```r
range(rd)
```

```
[1]  0.000 14.526
```

```r
library(pheatmap)
colnames(rd) <- paste0("m",1:5)
rownames(rd) <- paste0("m",1:5)
pheatmap(rd)
```

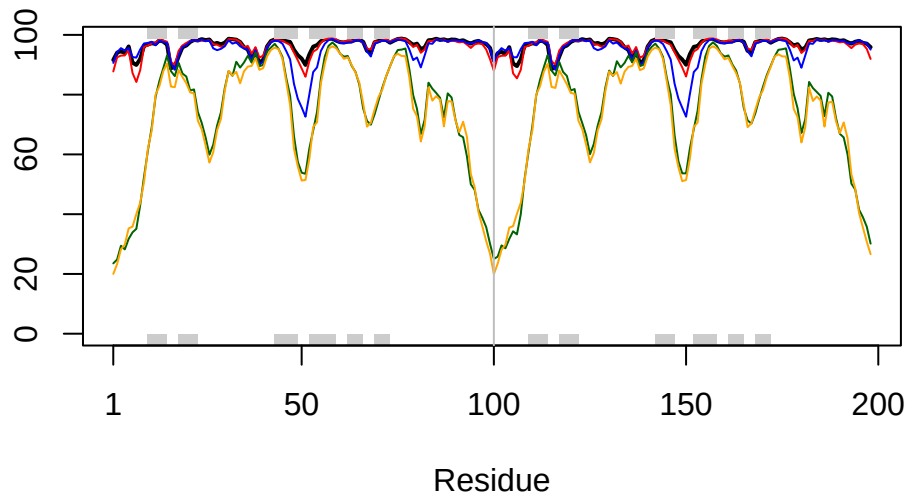


Now lets plot the pLDDT values across all models. Recall that this information is in the B-factor column of each model and that this is stored in our aligned pdbs object as `pdbs$b` with a row per structure/model.

```r
# Read a reference PDB structure
pdb <- read.pdb("1hsg")
```

```
  Note:  Accessing on-line PDB file
```

```r
plotb3(pdbs$b[1,], typ="l", lwd=2, sse=pdb)
points(pdbs$b[2,], typ="l", col="red")
points(pdbs$b[3,], typ="l", col="blue")
points(pdbs$b[4,], typ="l", col="darkgreen")
points(pdbs$b[5,], typ="l", col="orange")
abline(v=100, col="gray")
```



We can improve the superposition/fitting of our models by finding the most consistent "rigid core" common across all the models. For this we will use the core.find() function:

```r
core <- core.find(pdbs)
```

```
core size 197 of 198  vol = 5311.905
core size 196 of 198  vol = 4622.32
core size 195 of 198  vol = 1803.028
core size 194 of 198  vol = 1106.755
core size 193 of 198  vol = 1033.427
core size 192 of 198  vol = 985.198
core size 191 of 198  vol = 939.616
core size 190 of 198  vol = 896.766
core size 189 of 198  vol = 856.252
core size 188 of 198  vol = 822.829
```

```
core size 187 of 198  vol = 791.898
core size 186 of 198  vol = 762.687
core size 185 of 198  vol = 739.51
core size 184 of 198  vol = 712.268
core size 183 of 198  vol = 688.778
core size 182 of 198  vol = 669.224
core size 181 of 198  vol = 633.377
core size 180 of 198  vol = 614.667
core size 179 of 198  vol = 598.623
core size 178 of 198  vol = 582.74
core size 177 of 198  vol = 567.927
core size 176 of 198  vol = 553.536
core size 175 of 198  vol = 535.287
core size 174 of 198  vol = 522.628
core size 173 of 198  vol = 493.693
core size 172 of 198  vol = 479.665
core size 171 of 198  vol = 462.315
core size 170 of 198  vol = 448.76
core size 169 of 198  vol = 433.241
core size 168 of 198  vol = 420.053
core size 167 of 198  vol = 410.378
core size 166 of 198  vol = 398.085
core size 165 of 198  vol = 387.24
core size 164 of 198  vol = 373.948
core size 163 of 198  vol = 362.751
core size 162 of 198  vol = 348.611
core size 161 of 198  vol = 336.713
core size 160 of 198  vol = 324.778
core size 159 of 198  vol = 314.115
core size 158 of 198  vol = 302.268
core size 157 of 198  vol = 291.431
core size 156 of 198  vol = 280.553
core size 155 of 198  vol = 271.537
core size 154 of 198  vol = 262.651
core size 153 of 198  vol = 253.338
core size 152 of 198  vol = 244.266
core size 151 of 198  vol = 235.874
core size 150 of 198  vol = 228.071
core size 149 of 198  vol = 216.409
core size 148 of 198  vol = 203.865
core size 147 of 198  vol = 193.565
core size 146 of 198  vol = 185.893
core size 145 of 198  vol = 178.463
```

```
core size 144 of 198  vol = 170.348
core size 143 of 198  vol = 163.308
core size 142 of 198  vol = 153.436
core size 141 of 198  vol = 143.699
core size 140 of 198  vol = 137.681
core size 139 of 198  vol = 132.232
core size 138 of 198  vol = 124.684
core size 137 of 198  vol = 117.401
core size 136 of 198  vol = 109.876
core size 135 of 198  vol = 104.889
core size 134 of 198  vol = 98.885
core size 133 of 198  vol = 95.016
core size 132 of 198  vol = 91.683
core size 131 of 198  vol = 87.816
core size 130 of 198  vol = 84.046
core size 129 of 198  vol = 79.701
core size 128 of 198  vol = 76.896
core size 127 of 198  vol = 73.948
core size 126 of 198  vol = 70.293
core size 125 of 198  vol = 66.763
core size 124 of 198  vol = 64.12
core size 123 of 198  vol = 60.644
core size 122 of 198  vol = 57.631
core size 121 of 198  vol = 53.838
core size 120 of 198  vol = 50.347
core size 119 of 198  vol = 46.207
core size 118 of 198  vol = 43.507
core size 117 of 198  vol = 39.709
core size 116 of 198  vol = 36.796
core size 115 of 198  vol = 34.65
core size 114 of 198  vol = 31.443
core size 113 of 198  vol = 28.79
core size 112 of 198  vol = 26.627
core size 111 of 198  vol = 24.28
core size 110 of 198  vol = 22.088
core size 109 of 198  vol = 20.127
core size 108 of 198  vol = 18.254
core size 107 of 198  vol = 16.785
core size 106 of 198  vol = 15.453
core size 105 of 198  vol = 14.137
core size 104 of 198  vol = 12.84
core size 103 of 198  vol = 11.04
core size 102 of 198  vol = 10.012
```

```
core size 101 of 198  vol = 9.055
core size 100 of 198  vol = 8.017
core size 99 of 198  vol = 7.503
core size 98 of 198  vol = 6.268
core size 97 of 198  vol = 5.279
core size 96 of 198  vol = 4.152
core size 95 of 198  vol = 3.452
core size 94 of 198  vol = 2.854
core size 93 of 198  vol = 2.266
core size 92 of 198  vol = 1.944
core size 91 of 198  vol = 1.623
core size 90 of 198  vol = 1.382
core size 89 of 198  vol = 1.063
core size 88 of 198  vol = 0.873
core size 87 of 198  vol = 0.697
core size 86 of 198  vol = 0.552
core size 85 of 198  vol = 0.442
FINISHED: Min vol ( 0.5 ) reached
```

We can now use the identified core atom positions as a basis for a more suitable superposition and write out the fitted structures to a directory called corefit_structures:

```r
core.inds <- print(core, vol=0.5)
```

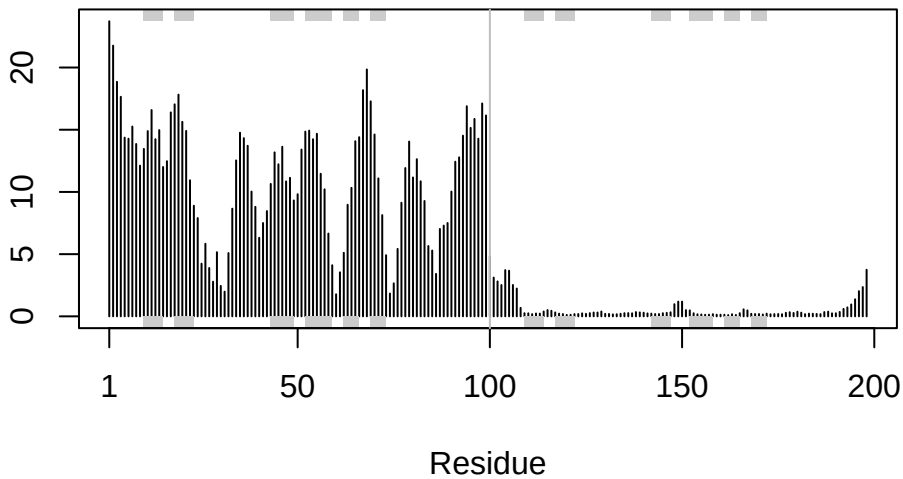
```
# 86 positions (cumulative volume <= 0.5 Angstrom^3)
  start end length
1     9  50     42
2    52  95     44
```

```r
xyz <- pdbfit(pdbs, core.inds, outpath="corefit_structures")
```

Now we can examine the RMSF between positions of the structure. RMSF is an often used measure of conformational variance along the structure

```r
rf <- rmsf(xyz)

plotb3(rf, sse=pdb)
abline(v=100, col="gray", ylab="RMSF")
```



Predicted Alignment Error for domains

Independent of the 3D structure, AlphaFold produces an output called Predicted Aligned Error (PAE). This is detailed in the JSON format result files, one for each model structure.

Below we read these files and see that AlphaFold produces a useful inter-domain prediction for model 1 (and 2) but not for model 5 (or indeed models 3, 4, and 5):

```
library(jsonlite)
results_dir <- "HIVpr_23119"

# Listing of all PAE JSON files
pae_files <- list.files(path=results_dir,
                        pattern=".*model.*\\.json",
                        full.names = TRUE)
```

```
pae1 <- read_json(pae_files[1],simplifyVector = TRUE)
pae5 <- read_json(pae_files[5],simplifyVector = TRUE)

attributes(pae1)
```

```
$names
```

```
[1] "plddt"   "max_pae" "pae"     "ptm"     "iptm"
```

```r
# Per-residue pLDDT scores
#  same as B-factor of PDB..
head(pae1$plddt)
```

```
[1] 91.62 94.06 94.56 93.88 96.12 90.69
```

The maximum PAE values are useful for ranking models. Here we can see that model 5 is much worse than model 1. The lower the PAE score the better. How about the other models, what are thir max PAE scores?

```r
pae1$max_pae
```

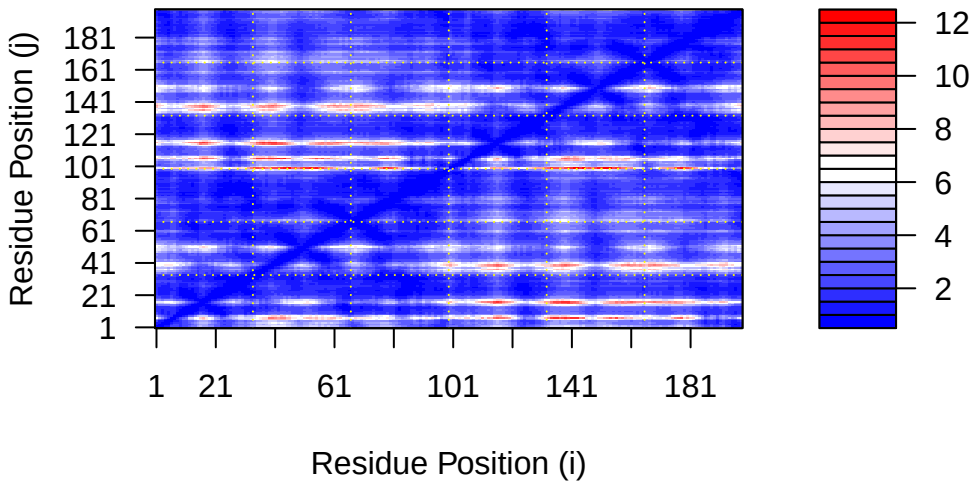
```
[1] 12.33594
```

```r
pae5$max_pae
```

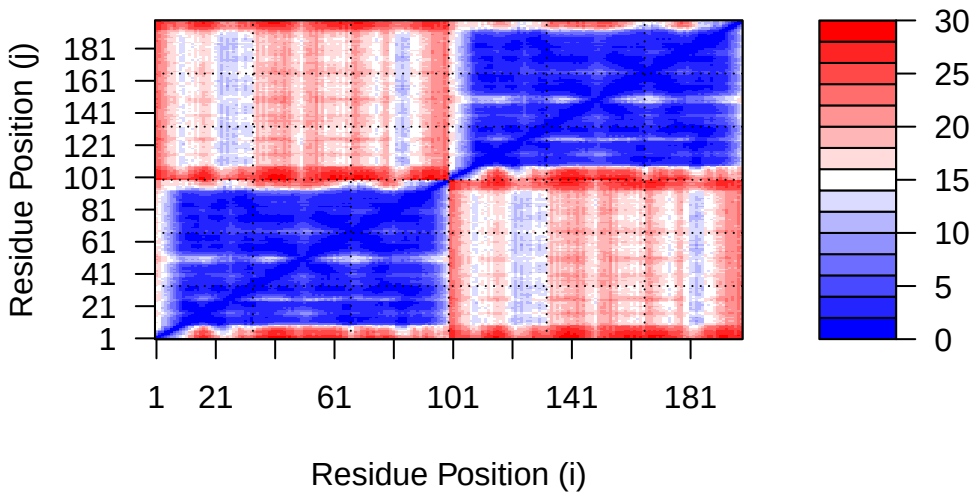
```
[1] 29.45312
```

We can plot the N by N (where N is the number of residues) PAE scores with ggplot or with functions from the Bio3D package:

```r
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)")
```

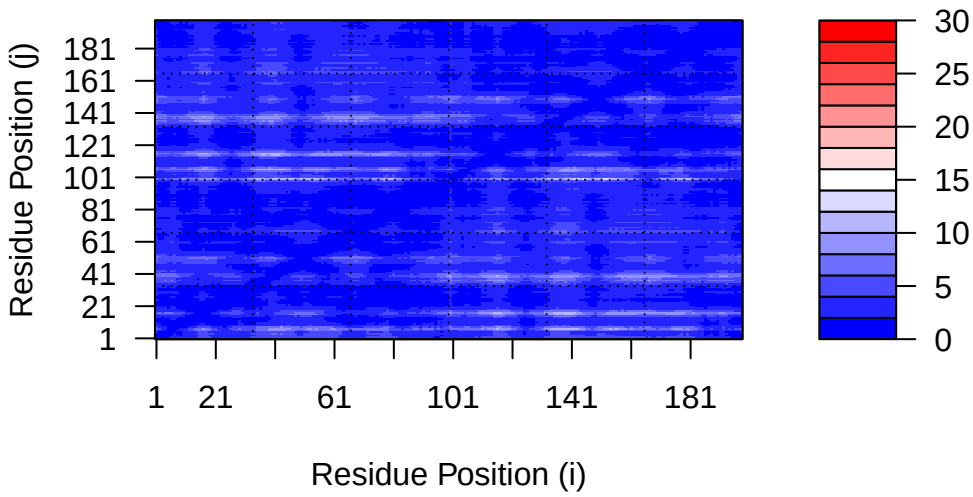



```
plot.dmat(pae5$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```



We should really plot all of these using the same z range. Here is the model 1 plot again but this time using the same data range as the plot for model 5:

```
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```



Residue conservation from alignment file

```
aln_file <- list.files(path=results_dir,
                       pattern=".a3m$",
                        full.names = TRUE)
aln_file
```

```
[1] "HIVpr_23119/HIVpr_23119.a3m"
```

```
aln <- read.fasta(aln_file[1], to.upper = TRUE)
```

```
[1] " ** Duplicated sequence id's: 101 **"
[2] " ** Duplicated sequence id's: 101 **"
```

How many sequences are in this alignment

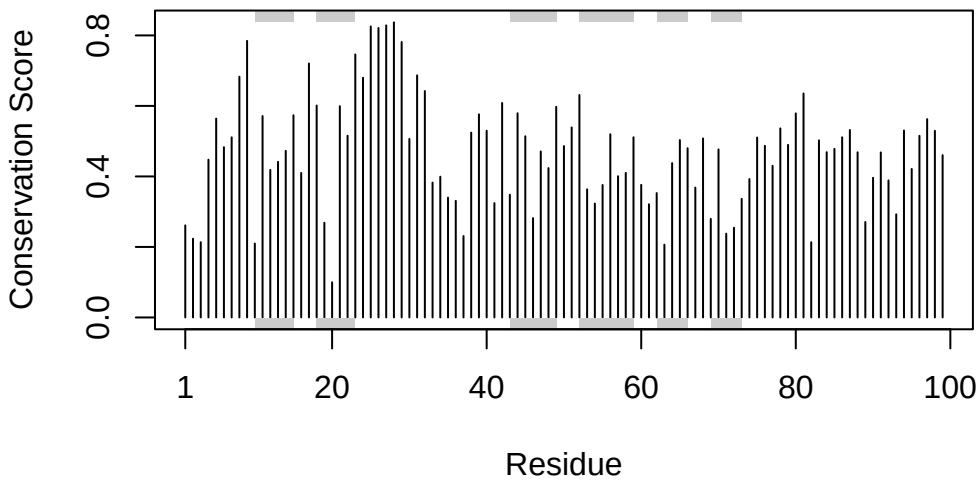
```
dim(aln$ali)
```

```
[1] 5397  132
```

We can score residue conservation in the alignment with the conserv() function.

```
sim <- conserv(aln)

plotb3(sim[1:99], sse=trim.pdb(pdb, chain="A"),
       ylab="Conservation Score")
```



Note the conserved Active Site residues D25, T26, G27, A28. These positions will stand out if we generate a consensus sequence with a high cutoff value:

```
con <- consensus(aln, cutoff = 0.9)
con$seq
```

```
  [1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [19] "-" "-" "-" "-" "-" "-" "D" "T" "G" "A" "-" "-" "-" "-" "-" "-" "-" "-"
 [37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-"
```

For a final visualization of these functionally important sites we can map this conservation score to the Occupancy column of a PDB file for viewing in molecular viewer programs such as Mol*, PyMol, VMD, chimera etc

```
m1.pdb <- read.pdb(pdb_files[1])
occ <- vec2resno(c(sim[1:99], sim[1:99]), m1.pdb$atom$resno)
write.pdb(m1.pdb, o=occ, file="m1_conserv.pdb")
```